

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: data = pd.read_csv(r"C:\\Users\\LORD\\Downloads\\flights.csv")
pd.set_option('display.max.columns' , 31)
```

```
C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\81526140.py:1: DtypeWarning: Columns (7,8) have mixed types. Specify d
type option on import or set low_memory=False.
  data = pd.read_csv(r"C:\\Users\\LORD\\Downloads\\flights.csv")
```

```
In [3]: airlines = pd.read_csv(r"C:\\Users\\LORD\\Downloads\\airlines.csv")
airlines.head()
```

```
Out[3]:
```

	IATA_CODE	AIRLINE
0	UA	United Air Lines Inc.
1	AA	American Airlines Inc.
2	US	US Airways Inc.
3	F9	Frontier Airlines Inc.
4	B6	JetBlue Airways

```
In [4]: airports = pd.read_csv(r"C:\\Users\\LORD\\Downloads\\airports.csv")
airports.head()
```

Out[4]:

	IATA_CODE	AIRPORT	CITY	STATE	COUNTRY	LATITUDE	LONGITUDE
0	ABE	Lehigh Valley International Airport	Allentown	PA	USA	40.65236	-75.44040
1	ABI	Abilene Regional Airport	Abilene	TX	USA	32.41132	-99.68190
2	ABQ	Albuquerque International Sunport	Albuquerque	NM	USA	35.04022	-106.60919
3	ABR	Aberdeen Regional Airport	Aberdeen	SD	USA	45.44906	-98.42183
4	ABY	Southwest Georgia Regional Airport	Albany	GA	USA	31.53552	-84.19447

```
In [5]: data['Flight_date'] = (  
    data['YEAR'].astype(str) + '-' +  
    data['MONTH'].astype(str).str.zfill(2) + '-' +  
    data['DAY'].astype(str).str.zfill(2)  
)
```

```
In [6]: data = data.drop(columns = ['YEAR', 'MONTH', 'DAY'])
```

```
In [7]: data['Flight_date'] = pd.to_datetime(data['Flight_date'] )  
data
```

Out[7]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2359	
5819075	4	B6	745	N828JB	JFK	PSE	2359	
5819076	4	B6	1503	N913JB	JFK	SJU	2359	
5819077	4	B6	333	N527JB	MCO	SJU	2359	
5819078	4	B6	839	N534JB	JFK	BQN	2359	

5819079 rows × 29 columns

In [8]:

```
data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5819079 entries, 0 to 5819078
Data columns (total 29 columns):
 #   Column                Dtype
---  -
 0   DAY_OF_WEEK           int64
 1   AIRLINE               object
 2   FLIGHT_NUMBER         int64
 3   TAIL_NUMBER           object
 4   ORIGIN_AIRPORT        object
 5   DESTINATION_AIRPORT   object
 6   SCHEDULED_DEPARTURE   int64
 7   DEPARTURE_TIME        float64
 8   DEPARTURE_DELAY       float64
 9   TAXI_OUT              float64
10  WHEELS_OFF            float64
11  SCHEDULED_TIME        float64
12  ELAPSED_TIME           float64
13  AIR_TIME              float64
14  DISTANCE              int64
15  WHEELS_ON             float64
16  TAXI_IN               float64
17  SCHEDULED_ARRIVAL     int64
18  ARRIVAL_TIME          float64
19  ARRIVAL_DELAY         float64
20  DIVERTED              int64
21  CANCELLED             int64
22  CANCELLATION_REASON   object
23  AIR_SYSTEM_DELAY      float64
24  SECURITY_DELAY        float64
25  AIRLINE_DELAY         float64
26  LATE_AIRCRAFT_DELAY   float64
27  WEATHER_DELAY         float64
28  Flight_date           datetime64[ns]
dtypes: datetime64[ns](1), float64(16), int64(7), object(5)
memory usage: 1.3+ GB

```

```

In [8]: data = data.drop_duplicates(subset = ['DAY_OF_WEEK', 'AIRLINE', 'FLIGHT_NUMBER', 'TAIL_NUMBER',
'ORIGIN_AIRPORT', 'DESTINATION_AIRPORT', 'SCHEDULED_DEPARTURE',

```

```
'DEPARTURE_TIME', 'DEPARTURE_DELAY', 'TAXI_OUT', 'WHEELS_OFF', 'SCHEDULED_TIME',
'ELAPSED_TIME', 'AIR_TIME', 'DISTANCE', 'WHEELS_ON', 'TAXI_IN', 'SCHEDULED_ARRIVAL',
'ARRIVAL_TIME', 'ARRIVAL_DELAY', 'DIVERTED', 'CANCELLED', 'CANCELLATION_REASON',
'AIR_SYSTEM_DELAY', 'SECURITY_DELAY']])
```

In [9]: `data.dropna(how = 'all')`

Out[9]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPA
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
...	...	...	...	...	...	...	...	
5819074	4	B6	688	N657JB	LAX	BOS	2359	
5819075	4	B6	745	N828JB	JFK	PSE	2359	
5819076	4	B6	1503	N913JB	JFK	SJU	2359	
5819077	4	B6	333	N527JB	MCO	SJU	2359	
5819078	4	B6	839	N534JB	JFK	BQN	2359	

5817999 rows × 29 columns

In [10]: `data['CANCELLATION_REASON'].fillna(0, inplace = True)`
`data`

```
C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\2408022044.py:1: SettingWithCopyWarning:
```

```
A value is trying to be set on a copy of a slice from a DataFrame
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
```

```
data['CANCELLATION_REASON'].fillna(0 , inplace = True)
```

Out[10]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_DELAY
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2359	
5819075	4	B6	745	N828JB	JFK	PSE	2359	
5819076	4	B6	1503	N913JB	JFK	SJU	2359	
5819077	4	B6	333	N527JB	MCO	SJU	2359	
5819078	4	B6	839	N534JB	JFK	BQN	2359	

5817999 rows × 9 columns

In [11]:

```
data['DEPARTURE_TIME'] = data['DEPARTURE_TIME'].astype('Int64')
data['DEPARTURE_DELAY'] = data['DEPARTURE_DELAY'].astype('Int64')
data
```

```

C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\1425546992.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
    data['DEPARTURE_TIME'] = data['DEPARTURE_TIME'].astype('Int64')
C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\1425546992.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
    data['DEPARTURE_DELAY'] = data['DEPARTURE_DELAY'].astype('Int64')

```

```

Out[11]:

```

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPA
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2359	
5819075	4	B6	745	N828JB	JFK	PSE	2359	
5819076	4	B6	1503	N913JB	JFK	SJU	2359	
5819077	4	B6	333	N527JB	MCO	SJU	2359	
5819078	4	B6	839	N534JB	JFK	BQN	2359	

5817999 rows × 29 columns

```

In [12]: data['SCHEDULED_DEPARTURE_INT'] = data['SCHEDULED_DEPARTURE'].fillna(0).astype(int)
data['hour'] = data['SCHEDULED_DEPARTURE_INT'] // 100

```

```
data['minute'] = data['SCHEDULED_DEPARTURE_INT'] % 100
data.loc[(data['hour'] > 23) | (data['minute'] > 59), ['hour', 'minute']] = np.nan
data['SCHEDULED_DEPARTURE_TIME'] = pd.to_datetime(data['hour'].astype('Int64').astype(str).str.zfill(2)
+ ':' +
data['minute'].astype('Int64').astype(str).str.zfill(2), format='%H:%M', errors='coerce').dt.time
```

C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\658690332.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
data['SCHEDULED_DEPARTURE_INT'] = data['SCHEDULED_DEPARTURE'].fillna(0).astype(int)
```

C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\658690332.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
data['hour'] = data['SCHEDULED_DEPARTURE_INT'] // 100
```

C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\658690332.py:3: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
data['minute'] = data['SCHEDULED_DEPARTURE_INT'] % 100
```

C:\Users\LORD\AppData\Local\Temp\ipykernel_18344\658690332.py:5: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
data['SCHEDULED_DEPARTURE_TIME'] = pd.to_datetime(data['hour'].astype('Int64').astype(str).str.zfill(2) + ':' +
```

In [20]:

```
data = data.drop(columns = ['SCHEDULED_DEPARTURE_INT'])
data
```


Out[20]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2359	
5819075	4	B6	745	N828JB	JFK	PSE	2359	
5819076	4	B6	1503	N913JB	JFK	SJU	2359	
5819077	4	B6	333	N527JB	MCO	SJU	2359	
5819078	4	B6	839	N534JB	JFK	BQN	2359	

5817999 rows × 29 columns

In [21]:

```
time_columns = ['SCHEDULED_DEPARTURE', 'DEPARTURE_TIME', 'WHEELS_OFF', 'WHEELS_ON', 'SCHEDULED_ARRIVAL',
                'ARRIVAL_TIME']

def convert_to_time(value) :
    if pd.isna(value) :
        return None
    value = int(value)
    hour = value // 100
    minute = value % 100
    return f"{hour:02d}:{minute:02d}"

for col in time_columns :
```

```
new_col = col + '_time'
data[new_col] = data[col].apply(convert_to_time)
```

Out[21]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2359	
5819075	4	B6	745	N828JB	JFK	PSE	2359	
5819076	4	B6	1503	N913JB	JFK	SJU	2359	
5819077	4	B6	333	N527JB	MCO	SJU	2359	
5819078	4	B6	839	N534JB	JFK	BQN	2359	

5817999 rows × 35 columns

In [26]:

```
pd.set_option('display.max.columns',35)
data.head(34)
```

Out[26]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTUR
0	4	AS	98	N407AS	ANC	SEA	5	
1	4	AA	2336	N3KUAA	LAX	PBI	10	
2	4	US	840	N171US	SFO	CLT	20	
3	4	AA	258	N3HYAA	LAX	MIA	20	
4	4	AS	135	N527AS	SEA	ANC	25	
5	4	DL	806	N3730B	SFO	MSP	25	
6	4	NK	612	N635NK	LAS	MSP	25	
7	4	US	2013	N584UW	LAX	CLT	30	
8	4	AA	1112	N3LAAA	SFO	DFW	30	
9	4	DL	1173	N826DN	LAS	ATL	30	
10	4	DL	2336	N958DN	DEN	ATL	30	
11	4	AA	1674	N853AA	LAS	MIA	35	
12	4	DL	1434	N547US	LAX	MSP	35	
13	4	DL	2324	N3751B	SLC	ATL	40	
14	4	DL	2440	N651DL	SEA	MSP	40	
15	4	AS	108	N309AS	ANC	SEA	45	
16	4	DL	1560	N3743H	ANC	SEA	45	
17	4	UA	1197	N78448	SFO	IAH	48	
18	4	AS	122	N413AS	ANC	PDX	50	
19	4	DL	1670	N806DN	PDX	MSP	50	
20	4	NK	520	N525NK	LAS	MCI	55	
21	4	AA	371	N3GXAA	SEA	MIA	100	
22	4	NK	214	N632NK	LAS	DFW	103	
23	4	AA	115	N3CTAA	LAX	MIA	105	
24	4	DL	1450	N671DN	LAS	MSP	105	

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTUR
25	4	UA	1545	N76517	LAX	IAH	115	
26	4	AS	130	N457AS	FAI	SEA	115	
27	4	NK	597	N528NK	MSP	FLL	115	
28	4	US	413	N571UW	LAS	CLT	120	
29	4	AA	2392	N3HRAA	DEN	MIA	120	
30	4	NK	168	N629NK	PHX	ORD	125	
31	4	AA	2211	N3CGAA	PHX	MIA	127	
32	4	AS	136	N431AS	ANC	SEA	135	
33	4	DL	95	N320US	SLC	ATL	140	

```
In [ ]: SCHEDULED_DEPARTURE    DEPARTURE_TIME    DEPARTURE_DELAY    TAXI_OUT            WHEELS_OFF            SCHEDULED_TIME
ELAPSED_TIME    AIR_TIME            DISTANCE            WHEELS_ON            TAXI_IN    SCHEDULED_ARRIVAL
ARRIVAL_TIME    ARRIVAL_DELAY
'SCHEDULED_DEPARTURE', 'DEPARTURE_TIME', 'WHEELS_OFF', 'WHEELS_ON', 'SCHEDULED_ARRIVAL', 'ARRIVAL_TIME'
```

```
In [27]: data = data.drop(columns = ['SCHEDULED_DEPARTURE', 'DEPARTURE_TIME', 'WHEELS_OFF', 'WHEELS_ON',
'SCHEDULED_ARRIVAL', 'ARRIVAL_TIME'])
```

Out[27]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DEPARTURE_DELAY	TAXI_OUT
0	4	AS	98	N407AS	ANC	SEA	-11	21.0
1	4	AA	2336	N3KUAA	LAX	PBI	-8	12.0
2	4	US	840	N171US	SFO	CLT	-2	16.0
3	4	AA	258	N3HYAA	LAX	MIA	-5	15.0
4	4	AS	135	N527AS	SEA	ANC	-1	11.0
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	-4	22.0
5819075	4	B6	745	N828JB	JFK	PSE	-4	17.0
5819076	4	B6	1503	N913JB	JFK	SJU	-9	17.0
5819077	4	B6	333	N527JB	MCO	SJU	-6	10.0
5819078	4	B6	839	N534JB	JFK	BQN	15	14.0

5817999 rows × 29 columns

In [28]:

```
data.head(35)
```

Out[28]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DEPARTURE_DELAY	TAXI_OUT	SCHI
0	4	AS	98	N407AS	ANC	SEA	-11	21.0	
1	4	AA	2336	N3KUAA	LAX	PBI	-8	12.0	
2	4	US	840	N171US	SFO	CLT	-2	16.0	
3	4	AA	258	N3HYAA	LAX	MIA	-5	15.0	
4	4	AS	135	N527AS	SEA	ANC	-1	11.0	
5	4	DL	806	N3730B	SFO	MSP	-5	18.0	
6	4	NK	612	N635NK	LAS	MSP	-6	11.0	
7	4	US	2013	N584UW	LAX	CLT	14	13.0	
8	4	AA	1112	N3LAAA	SFO	DFW	-11	17.0	
9	4	DL	1173	N826DN	LAS	ATL	3	12.0	
10	4	DL	2336	N958DN	DEN	ATL	-6	12.0	
11	4	AA	1674	N853AA	LAS	MIA	-8	21.0	
12	4	DL	1434	N547US	LAX	MSP	0	18.0	
13	4	DL	2324	N3751B	SLC	ATL	-6	18.0	
14	4	DL	2440	N651DL	SEA	MSP	-1	28.0	
15	4	AS	108	N309AS	ANC	SEA	-4	17.0	
16	4	DL	1560	N3743H	ANC	SEA	-14	25.0	
17	4	UA	1197	N78448	SFO	IAH	-6	11.0	
18	4	AS	122	N413AS	ANC	PDX	-4	11.0	
19	4	DL	1670	N806DN	PDX	MSP	-5	9.0	
20	4	NK	520	N525NK	LAS	MCI	25	11.0	
21	4	AA	371	N3GXAA	SEA	MIA	-8	30.0	
22	4	NK	214	N632NK	LAS	DFW	-1	13.0	
23	4	AA	115	N3CTAA	LAX	MIA	-2	14.0	
24	4	DL	1450	N671DN	LAS	MSP	-3	11.0	

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DEPARTURE_DELAY	TAXI_OUT	SCHI
25	4	UA	1545	N76517	LAX	IAH	-3	11.0	
26	4	AS	130	N457AS	FAI	SEA	-8	25.0	
27	4	NK	597	N528NK	MSP	FLL	12	14.0	
28	4	US	413	N571UW	LAS	CLT	-10	12.0	
29	4	AA	2392	N3HRAA	DEN	MIA	21	12.0	
30	4	NK	168	N629NK	PHX	ORD	72	9.0	
31	4	AA	2211	N3CGAA	PHX	MIA	-11	10.0	
32	4	AS	136	N431AS	ANC	SEA	<NA>	NaN	
33	4	DL	95	N320US	SLC	ATL	-6	43.0	
34	4	NK	298	N514NK	LAS	IAH	-4	10.0	

```
In [ ]: DEPARTURE_DELAY TAXI_OUT          SCHEDULED_TIME ELAPSED_TIME   AIR_TIME      DISTANCE      TAXI_IN
ARRIVAL_DELAY
```

```
In [36]: columns =
['DEPARTURE_DELAY', 'TAXI_OUT', 'SCHEDULED_TIME', 'ELAPSED_TIME', 'AIR_TIME', 'TAXI_IN', 'ARRIVAL_DELAY']

for i in columns :
    data[i + '_new'] = data[i]
```

Out[36]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DEPARTURE_DELAY	TAXI_OUT
0	4	AS	98	N407AS	ANC	SEA	-11	21.0
1	4	AA	2336	N3KUAA	LAX	PBI	-8	12.0
2	4	US	840	N171US	SFO	CLT	-2	16.0
3	4	AA	258	N3HYAA	LAX	MIA	-5	15.0
4	4	AS	135	N527AS	SEA	ANC	-1	11.0
...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	-4	22.0
5819075	4	B6	745	N828JB	JFK	PSE	-4	17.0
5819076	4	B6	1503	N913JB	JFK	SJU	-9	17.0
5819077	4	B6	333	N527JB	MCO	SJU	-6	10.0
5819078	4	B6	839	N534JB	JFK	BQN	15	14.0

5817999 rows × 36 columns

In []:

```
data = data.drop(columns =  
['DEPARTURE_DELAY', 'TAXI_OUT', 'SCHEDULED_TIME', 'ELAPSED_TIME', 'AIR_TIME', 'TAXI_IN', 'ARRIVAL_DELAY'])
```

In [58]:

```
data.head(35)
```


Out[58]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANCELLED
0	4	AS	98	N407AS	ANC	SEA	1448	0	0
1	4	AA	2336	N3KUAA	LAX	PBI	2330	0	0
2	4	US	840	N171US	SFO	CLT	2296	0	0
3	4	AA	258	N3HYAA	LAX	MIA	2342	0	0
4	4	AS	135	N527AS	SEA	ANC	1448	0	0
5	4	DL	806	N3730B	SFO	MSP	1589	0	0
6	4	NK	612	N635NK	LAS	MSP	1299	0	0
7	4	US	2013	N584UW	LAX	CLT	2125	0	0
8	4	AA	1112	N3LAAA	SFO	DFW	1464	0	0
9	4	DL	1173	N826DN	LAS	ATL	1747	0	0
10	4	DL	2336	N958DN	DEN	ATL	1199	0	0
11	4	AA	1674	N853AA	LAS	MIA	2174	0	0
12	4	DL	1434	N547US	LAX	MSP	1535	0	0
13	4	DL	2324	N3751B	SLC	ATL	1590	0	0
14	4	DL	2440	N651DL	SEA	MSP	1399	0	0
15	4	AS	108	N309AS	ANC	SEA	1448	0	0
16	4	DL	1560	N3743H	ANC	SEA	1448	0	0
17	4	UA	1197	N78448	SFO	IAH	1635	0	0
18	4	AS	122	N413AS	ANC	PDX	1542	0	0
19	4	DL	1670	N806DN	PDX	MSP	1426	0	0
20	4	NK	520	N525NK	LAS	MCI	1139	0	0
21	4	AA	371	N3GXAA	SEA	MIA	2724	0	0
22	4	NK	214	N632NK	LAS	DFW	1055	0	0
23	4	AA	115	N3CTAA	LAX	MIA	2342	0	0
24	4	DL	1450	N671DN	LAS	MSP	1299	0	0

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANCELLED
25	4	UA	1545	N76517	LAX	IAH	1379	0	0
26	4	AS	130	N457AS	FAI	SEA	1533	0	0
27	4	NK	597	N528NK	MSP	FLL	1487	0	0
28	4	US	413	N571UW	LAS	CLT	1916	0	0
29	4	AA	2392	N3HRAA	DEN	MIA	1709	0	0
30	4	NK	168	N629NK	PHX	ORD	1440	0	0
31	4	AA	2211	N3CGAA	PHX	MIA	1972	0	0
32	4	AS	136	N431AS	ANC	SEA	1448	0	1
33	4	DL	95	N320US	SLC	ATL	1590	0	0
34	4	NK	298	N514NK	LAS	IAH	1222	0	0

```
In [ ]: DEPARTURE_DELAY_new    TAXI_OUT_new    SCHEDULED_TIME_new    ELAPSED_TIME_new    AIR_TIME_new
        TAXI_IN_new    ARRIVAL_DELAY_new
```

```
In [60]: float_columns =
        ['DEPARTURE_DELAY_new', 'TAXI_OUT_new', 'SCHEDULED_TIME_new', 'ELAPSED_TIME_new', 'AIR_TIME_new', 'TAXI_IN_new', 'ARRIVAL_DELAY_new']

        for a in float_columns :
            data[a] = data[a].astype('Int64').round()

        data
```

Out[60]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANCELLED
0	4	AS	98	N407AS	ANC	SEA	1448	0	
1	4	AA	2336	N3KUAA	LAX	PBI	2330	0	
2	4	US	840	N171US	SFO	CLT	2296	0	
3	4	AA	258	N3HYAA	LAX	MIA	2342	0	
4	4	AS	135	N527AS	SEA	ANC	1448	0	
...	...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2611	0	
5819075	4	B6	745	N828JB	JFK	PSE	1617	0	
5819076	4	B6	1503	N913JB	JFK	SJU	1598	0	
5819077	4	B6	333	N527JB	MCO	SJU	1189	0	
5819078	4	B6	839	N534JB	JFK	BQN	1576	0	

5817999 rows × 30 columns

In [57]:

```
data['Cancellation_Reason_Desc'] = data['CANCELLATION_REASON'].apply(lambda x :  
'Career_Airline_related_issue' if x == 'A' else 'Weather_related_cancellation' if x == 'B' else  
'Security_related_cancellation' if x == 'C' else 'No_cancellation')  
  
data
```

Out[57]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANCEL
0	4	AS	98	N407AS	ANC	SEA	1448	0	
1	4	AA	2336	N3KUAA	LAX	PBI	2330	0	
2	4	US	840	N171US	SFO	CLT	2296	0	
3	4	AA	258	N3HYAA	LAX	MIA	2342	0	
4	4	AS	135	N527AS	SEA	ANC	1448	0	
...	...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2611	0	
5819075	4	B6	745	N828JB	JFK	PSE	1617	0	
5819076	4	B6	1503	N913JB	JFK	SJU	1598	0	
5819077	4	B6	333	N527JB	MCO	SJU	1189	0	
5819078	4	B6	839	N534JB	JFK	BQN	1576	0	

5817999 rows × 30 columns

In [63]:

```
null_column_to_zero =  
['AIR_SYSTEM_DELAY', 'SECURITY_DELAY', 'AIRLINE_DELAY', 'LATE_AIRCRAFT_DELAY', 'WEATHER_DELAY']  
  
null_columns_to_avg =  
['DEPARTURE_DELAY_new', 'TAXI_OUT_new', 'SCHEDULED_TIME_new', 'ELAPSED_TIME_new', 'AIR_TIME_new', 'TAXI_IN_new']  
  
for i in null_column_to_zero :  
    data[i] = data[i].fillna(0)  
  
for i in null_columns_to_avg :  
    data[i] = data[i].fillna(int(data[i].mean()))
```

```
data
```

Out[63]:

	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANCEL
0	4	AS	98	N407AS	ANC	SEA	1448	0	
1	4	AA	2336	N3KUAA	LAX	PBI	2330	0	
2	4	US	840	N171US	SFO	CLT	2296	0	
3	4	AA	258	N3HYAA	LAX	MIA	2342	0	
4	4	AS	135	N527AS	SEA	ANC	1448	0	
...	...	...	...	...	...	...	...	...	...
5819074	4	B6	688	N657JB	LAX	BOS	2611	0	
5819075	4	B6	745	N828JB	JFK	PSE	1617	0	
5819076	4	B6	1503	N913JB	JFK	SJU	1598	0	
5819077	4	B6	333	N527JB	MCO	SJU	1189	0	
5819078	4	B6	839	N534JB	JFK	BQN	1576	0	

5817999 rows × 30 columns

```
In [65]: data2 = pd.merge(left = data , right = airlines , left_on = 'AIRLINE' , right_on = 'IATA_CODE')
data2 = data2.drop(columns = ['IATA_CODE'])
data2
```

Out[65]:

	DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
--	-------------	-----------	---------------	-------------	----------------	---------------------	----------	----------	------

0	4	AS	98	N407AS	ANC	SEA	1448	0
1	4	AS	135	N527AS	SEA	ANC	1448	0
2	4	AS	108	N309AS	ANC	SEA	1448	0
3	4	AS	122	N413AS	ANC	PDX	1542	0
4	4	AS	130	N457AS	FAI	SEA	1533	0
...	...	...	...	...	...	...	...	...
5817994	4	VX	769	N622VA	LGA	DAL	1381	0
5817995	4	VX	357	N284VA	BOS	SFO	2704	0
5817996	4	VX	1916	N853VA	SFO	LAS	414	0
5817997	4	VX	490	N840VA	LAX	LAS	236	0
5817998	4	VX	48	N281VA	OGG	SFO	2338	0

5817999 rows × 31 columns

In []:

IATA_CODE	AIRPORT	CITY	STATE	COUNTRY	LATITUDE	LONGITUDE
-----------	---------	------	-------	---------	----------	-----------

In [67]:

```
data2 = data2.merge(airports,left_on = 'ORIGIN_AIRPORT' , right_on = 'IATA_CODE' , how = 'inner')  
data2 = data2.merge(airports,left_on = 'DESTINATION_AIRPORT' , right_on = 'IATA_CODE' , how = 'inner')
```

Out[67]:

	DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
0	4	AS	98	N407AS	ANC	SEA	1448	0	
1	4	AS	108	N309AS	ANC	SEA	1448	0	
2	4	AS	136	N431AS	ANC	SEA	1448	0	
3	4	AS	134	N464AS	ANC	SEA	1448	0	
4	4	AS	114	N303AS	ANC	SEA	1448	0	
...	...	...	...	...	...	...	...	...	...
5331841	2	EV	2734	N884AS	MEI	PIB	69	0	
5331842	3	EV	2730	N880AS	MEI	PIB	69	0	
5331843	3	EV	2734	N907EV	MEI	PIB	69	0	
5331844	4	EV	2730	N907EV	MEI	PIB	69	0	

DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
5331845	4	EV	2734	N907EV	MEI	PIB	69	0

5331846 rows × 52 columns

In [69]:

```
pd.set_option('display.max.columns',52)  
data2
```

Out[69]:

	DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
0	4	AS	98	N407AS	ANC	SEA	1448	0	
1	4	AS	108	N309AS	ANC	SEA	1448	0	
2	4	AS	136	N431AS	ANC	SEA	1448	0	
3	4	AS	134	N464AS	ANC	SEA	1448	0	
4	4	AS	114	N303AS	ANC	SEA	1448	0	
...	...	...	...	...	...	...	...	...	...
5331841	2	EV	2734	N884AS	MEI	PIB	69	0	
5331842	3	EV	2730	N880AS	MEI	PIB	69	0	
5331843	3	EV	2734	N907EV	MEI	PIB	69	0	
5331844	4	EV	2730	N907EV	MEI	PIB	69	0	

DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
5331845	4	EV	2734	N907EV	MEI	PIB	69	0

5331846 rows × 52 columns

In [71]:

```
final_data = data2.drop(columns =
['IATA_CODE_x', 'IATA_CODE_y', 'AIRPORT_y', 'CITY_y', 'STATE_y', 'COUNTRY_y', 'LATITUDE_y', 'LONGITUDE_y'])
final_data
```

Out[71]:

	DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
0	4	AS	98	N407AS	ANC	SEA	1448	0	
1	4	AS	108	N309AS	ANC	SEA	1448	0	
2	4	AS	136	N431AS	ANC	SEA	1448	0	
3	4	AS	134	N464AS	ANC	SEA	1448	0	
4	4	AS	114	N303AS	ANC	SEA	1448	0	
...	...	...	...	...	...	...	...	...	...
5331841	2	EV	2734	N884AS	MEI	PIB	69	0	
5331842	3	EV	2730	N880AS	MEI	PIB	69	0	
5331843	3	EV	2734	N907EV	MEI	PIB	69	0	
5331844	4	EV	2730	N907EV	MEI	PIB	69	0	

	DAY_OF_WEEK	AIRLINE_x	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	DISTANCE	DIVERTED	CANC
	5331845	4	EV	2734	N907EV	MEI	PIB	69	0

5331846 rows × 44 columns

In [72]:

```
final_data = final_data.drop(columns = ['IATA_CODE'])
final_data = final_data.rename(columns =
{'AIRLINE_x': 'AIRLINE_CODE', 'AIRLINE_y': 'AIRLINE', 'ORIGIN_AIRPORT': 'ORIGIN_AIRPORT_CODE', 'AIRPORT_x': 'ORIG
final_data
```

Out[72]:

	DAY_OF_WEEK	AIRLINE_CODE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT_CODE	DESTINATION_AIRPORT_CODE	DISTANCE
0	4	AS	98	N407AS	ANC	SEA	1448
1	4	AS	108	N309AS	ANC	SEA	1448
2	4	AS	136	N431AS	ANC	SEA	1448
3	4	AS	134	N464AS	ANC	SEA	1448
4	4	AS	114	N303AS	ANC	SEA	1448
...	...	...	...	...	...	...	...
5331841	2	EV	2734	N884AS	MEI	PIB	69
5331842	3	EV	2730	N880AS	MEI	PIB	69
5331843	3	EV	2734	N907EV	MEI	PIB	69
5331844	4	EV	2730	N907EV	MEI	PIB	69
5331845	4	EV	2734	N907EV	MEI	PIB	69

5331846 rows × 43 columns

```
In [73]: final_data.to_csv(r"C:\\Users\\LORD\\Downloads\\final_flights_data.csv")
```

```
In [ ]:
```

```
In [ ]:
```